Type MS · Explain how the eye focuses on near and distant objects

Answer 1 of 7 · Question 2 part b · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

2	(a)	(i)	• lens in correct position; • attached both ends to suspensory ligaments;	MP1- accept lens between ciliary muscles/diameter more than width of ciliary muscle MP2- accept suspensory ligament attached to ciliary muscle	2
		(ii)	A = cornea; B = retina; C = optic nerve;		3
	(b)		• ciliary muscles relax; • suspensory ligaments tighten/taut; • they pull on lens/lens stretched; • lens is thin(er)/flat(er/less convex); • less refraction/bending of (light)	Mp2- ignore suspensory ligament contract MP5= do not accept less light refracted	4
					Total = 9

Type MS · Explain how the eye focuses on near and distant objects

Answer 1 of 7 · Question 2 part b · 4 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how the eye focuses on near and distant objects

Answer 2 of 7 · Question 5 part a · 3 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

Question number				
5	(a)		*near* *distant* lens thick thin; circular ciliary muscle contracted relaxed; suspensory ligaments loose stretched;	3
	(b)		• two eyes; • each giving slightly different view; • allows better judgement of distances / depth;	3
	(c)		any six from • test with two eyes then one; • use same person/people of similar age/same eye condition; • (throw and) catch a ball; • 10 times/suitable repeats; • record number of catches • ref control variable e.g. lighting/wind condition/same size ball/throw from the same distance;	6

Total 12 marks

Type MS · Explain how the eye focuses on near and distant objects

Answer 2 of 7 · Question 5 part a · 3 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how the eye focuses on near and distant objects

Answer 3 of 7 · Question 4 part b.ii · 1 mark

2023 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
4	(a)	any two from • hearing/sound; • touch; • taste; • smell;		2
	(b) (i)	any two from • as thickness decreases, focal length increases; • interval between focal lengths increases as thickness decreases; • (constant when two variables multiplied hence) inversely proportional;		2
	(ii)	diameter decreases;		1
	(iii)	any four from • rays from near object diverging; • ciliary muscle contracts; • suspensory ligaments slacken/relax; • so lens becomes thicker/more convex; • more powerful lens/light refracted more/increased refraction; • (allows it to focus because) focal length becomes shorter/decrease;		4
	(c) (i)	between 575 and 580 nm;		1
	(ii)	• 400 - 530; • 130 nm;	ECF	2
	(iii)	• greater range of wavelengths can be seen/absorb light of different wavelengths; • gives better colour perception/vision/differentiation/can see different colours;		2

Total 14

Type MS · Explain how the eye focuses on near and distant objects

Answer 3 of 7 · Question 4 part b.ii · 1 mark

2023 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how the eye focuses on near and distant objects

Answer 4 of 7 · Question 4 part b.iii · 4 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
4	(a)	any two from • hearing/sound; • touch; • taste; • smell;		2
	(b) (i)	any two from • as thickness decreases, focal length increases; • interval between focal lengths increases as thickness decreases; • (constant when two variables multiplied hence) inversely proportional;		2
	(ii)	diameter decreases;		1
	(iii)	any four from • rays from near object diverging; • ciliary muscle contracts; • suspensory ligaments slacken/relax; • so lens becomes thicker/more convex; • more powerful lens/light refracted more/increased refraction; • (allows it to focus because) focal length becomes shorter/decrease;		4
	(c) (i)	between 575 and 580 nm;		1
	(ii)	• 400 - 530; • 130 nm;	ECF	2
	(iii)	• greater range of wavelengths can be seen/absorb light of different wavelengths; • gives better colour perception/vision/differentiation/can see different colours;		2

Total 14

Type MS · Explain how the eye focuses on near and distant objects

Answer 4 of 7 · Question 4 part b.iii · 4 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how the eye focuses on near and distant objects

Answer 5 of 7 · Question 3 part a.iii · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

3 (a)	(i)	Part W - pupil; Part X – retina;	1 mark for each correctly named part	2
	(ii)	• sends (electrical) impulses from photoreceptors/rods/cones; • to the brain/CNS; OR • sends (electrical) impulses from brain/CNS; • to iris/ciliary muscles;		2
	(iii)	lens becomes thinner/flatter/less convex; less refraction;	allow less light refracted	2
	(iv)	Any two from: • shape of cornea restored; • light refracted, onto the retina/correctly/notmally; • correct information sent to the brain; • image/vision not blurred/clear;		2
(b)		• measure the size/diameter of the pupil; • in a dark room; • shine a light into the eye; • measure/compare size/diameter of the pupil;	accept ref to different light intensities	4

Total Marks 12

Type MS · Explain how the eye focuses on near and distant objects

Answer 5 of 7 · Question 3 part a.iii · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how the eye focuses on near and distant objects

Answer 6 of 7 · Question 2 part a.i · 2 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

2	(a)	(i)	• lens in correct position; • attached both ends to suspensory ligaments;	MP1- accept lens between ciliary muscles/diameter more than width of ciliary muscle MP2- accept suspensory ligament attached to ciliary muscle	2
		(ii)	A = cornea; B = retina; C = optic nerve;		3
	(b)		• ciliary muscles relax; • suspensory ligaments tighten/taut; • they pull on lens/lens stretched; • lens is thin(er)/flat(er/less convex); • less refraction/bending of (light)	Mp2- ignore suspensory ligament contract MP5= do not accept less light refracted	4
					Total = 9

Type MS · Explain how the eye focuses on near and distant objects

Answer 6 of 7 · Question 2 part a.i · 2 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how the eye focuses on near and distant objects

Answer 7 of 7 · Question 6 part whole · 7 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

Number			
6	Any seven from the following: • cornea refracts light; • ciliary muscles contract/relax; • to change the shape of the lens; • (the lens) refracts light; • retina contains receptor/rod/cone cells; • (retina) converts light into electrical/nerve impulses; • optic nerve contains sensory neurones; • optic/sensory neurones carry impulses to the brain;	Allow signals as an alternative to impulses throughout Ignore shorten/widen lens	7

Total marks 7

Type MS · Explain how the eye focuses on near and distant objects

Answer 7 of 7 · Question 6 part whole · 7 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Answer	Notes	Marks